

COMP 2121 Formula Sheet

Rule of Inference	Related Logical Implication	Name of Rule
1) $\frac{p \quad p \rightarrow q}{\therefore q}$	$[p \wedge (p \rightarrow q)] \rightarrow q$	Rule of Detachment (Modus Ponens)
2) $\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Law of the Syllogism
3) $\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$	$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$	Modus Tollens
4) $\frac{p \quad q}{\therefore p \wedge q}$		Rule of Conjunction
5) $\frac{p \vee q \quad \neg p}{\therefore q}$	$[(p \vee q) \wedge \neg p] \rightarrow q$	Rule of Disjunctive Syllogism
6) $\frac{\neg p \rightarrow F_0}{\therefore p}$	$(\neg p \rightarrow F_0) \rightarrow p$	Rule of Contradiction
7) $\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Rule of Conjunctive Simplification
8) $\frac{p}{\therefore p \vee q}$	$p \rightarrow p \vee q$	Rule of Disjunctive Amplification
9) $\frac{p \wedge q \quad p \rightarrow (q \rightarrow r)}{\therefore r}$	$[(p \wedge q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow r$	Rule of Conditional Proof
10) $\frac{p \rightarrow r \quad q \rightarrow r}{\therefore (p \vee q) \rightarrow r}$	$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$	Rule for Proof by Cases
11) $\frac{p \rightarrow q \quad r \rightarrow s \quad p \vee r}{\therefore q \vee s}$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$	Rule of the Constructive Dilemma
12) $\frac{p \rightarrow q \quad r \rightarrow s \quad \neg q \vee \neg s}{\therefore \neg p \vee \neg r}$	$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)] \rightarrow (\neg p \vee \neg r)$	Rule of the Destructive Dilemma

No Name identity: $p \rightarrow q \Leftrightarrow \neg p \vee q$

Contrapositive: $p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$

Negation of Implication: $\neg(p \rightarrow q) \Leftrightarrow p \wedge (\neg q)$

Permutations/Combinations – without repetition:

$$P(n, r) = \frac{n!}{(n-r)!} \quad C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

Permutations/Combinations – with repetition:

$$P(n, r) = n^r \quad C(n+r-1, r) = \binom{n+r-1}{r}$$

Given n items with n_1 alike, n_2 alike, ..., n_k alike, the number of permutations of all n items is:

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Set Theory:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

The Laws of Logic

For any primitive statements p, q, r , any tautology T_0 , and any contradiction F_0 ,

1) $\neg\neg p \Leftrightarrow p$ Law of Double Negation

2) $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$ DeMorgan's Laws
 $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

3) $p \vee q \Leftrightarrow q \vee p$ Commutative Laws
 $p \wedge q \Leftrightarrow q \wedge p$

4) $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r^\dagger$ Associative Laws
 $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$

5) $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$ Distributive Laws
 $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

6) $p \vee p \Leftrightarrow p$ Idempotent Laws
 $p \wedge p \Leftrightarrow p$

7) $p \vee F_0 \Leftrightarrow p$ Identity Laws
 $p \wedge T_0 \Leftrightarrow p$

8) $p \vee \neg p \Leftrightarrow T_0$ Inverse Laws
 $p \wedge \neg p \Leftrightarrow F_0$

9) $p \vee T_0 \Leftrightarrow T_0$ Domination Laws
 $p \wedge F_0 \Leftrightarrow F_0$

10) $p \vee (p \wedge q) \Leftrightarrow p$ Absorption Laws
 $p \wedge (p \vee q) \Leftrightarrow p$